Expressing one number as a percentage of another

Task A: Writing any fraction as a percentage

1) Only shade in the equivalent percentages of the following fractions.

$\frac{1}{4} = 25\%$ $\frac{1}{5} = 20\%$

$\frac{9}{20} = 45\%$ $\frac{7}{100} = 7\%$

$\frac{11}{25} = 44\%$ $\frac{11}{20} = 55\%$

$\frac{1}{10} = 10\%$ $\frac{7}{25} = 28\%$

$\frac{31}{200} = 15.5\%$ $\frac{3}{20} = 15\%$

$\frac{15}{200} = 7.5\%$ $\frac{63}{100} = 63\%$

25%	20%	5%	31%	11%	15%
10%	20%	17%	5%	15.5%	17%
2%	8%	13%	7%	8%	5%
31%	4%	45%	14%	6%	8%
29%	28%	23%	11%	44%	55%
10%	21%	59%	3%	7.5%	63%

2) Convert the following fractions into a percentage. Give your answers correct to 1 decimal place if needed.

a) $\frac{3}{5}$

$$\frac{3}{5} = \frac{3 \times 20}{5 \times 20} = \frac{60}{100} = 60\%$$

$$5\overline{)3.{}^{3}0} = 0.6$$

$$0.6 = 60\%$$

b) $\frac{7}{20}$

$$\frac{7}{20} = \frac{7 \times 5}{20 \times 5} = \frac{35}{100} = 35\%$$

$$20\overline{)7\,.{}^{7}0{}^{10}0} = 0.35$$

$$0.35 = 35\%$$

c) $\frac{1}{8}$

$$8\overline{)1\,.{}^{1}0{}^{2}0{}^{4}0} = 0.125$$

$$\frac{1}{8} = 0.125$$

$$0.125 = 12.5\%$$

d) $\frac{1}{11}$

$$\frac{1}{11} = 1 \div 11$$

$$11\overline{)1\,{}^{1}0{}^{10}0{}^{1}0{}^{10}0} = 0\,0\,9\,0\,9\ldots$$

$$\frac{1}{11} = 0.\dot{0}\dot{9}$$

$$0.\dot{0}\dot{9} = 0.0909\ldots = 9.\dot{0}\dot{9} = 9.1\%$$

Task B: Writing any number as a percentage of another

1) Work out the following, giving your answer to 2 d.p. where appropriate:

a) 21 as a percentage of 25 84%

b) 60 as a percentage of 80 75%

c) 84 as a percentage of 240 35%

d) 81 as a percentage of 240 33.75%

e) 37 as a percentage of 60 61.67% (2 d.p.)

f) 19 as a percentage of 40 47.5%

2) Laura is ill and misses her first day of school.

a) What is her percentage attendance after day one? 0%

Laura is better the next day and attends school.

b) What is her percentage attendance after day two? 50%

Laura attends the rest of that week and all of the next week.

c) What is her percentage attendance after day five? 80%

d) What is her percentage attendance after day ten? 90%

Laura says that she will need to attend every day for the rest of the year to get 100% attendance.

e) Is Laura correct? Why or why not?

No, Laura is not correct – she cannot get 100% attendance for the year, as she has already missed a day.
The highest Laura's attendance can be is 99% (as $189 \div 190 \times 100 = 99.473684...$). The lowest Laura's attendance could be is 5% (as $9 \div 190 \times 100 = 4.736842...$)